# ISSAM ELGHBALI

Software Engineer | Data Science & AI

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GitHub · LinkedIn · Online Portfolio

## Professional Profile

Master's student in Data Science and Intelligent Systems at the Faculty of Sciences and Techniques of Fez, with hands-on experience in software engineering, full-stack applications, and applied machine learning. Develops RESTful APIs and layered Java/Spring Boot systems alongside Python-based analytics, explainable ML workflows, and interactive web applications. Project work spans enterprise resource management, fraud analysis, customer retention, and football recommendation systems, with emphasis on maintainability, evaluation, and practical integration.

## Education

**Master's Degree in Data Science & Intelligent Systems** 2025 – Present
Faculty of Sciences and Techniques of Fez (FST Fez), Morocco
Advanced studies in machine learning, artificial intelligence, and intelligent systems.

**Licence Sciences et Techniques (LST) – Computer Engineering** 2024 – 2025
Faculty of Sciences and Techniques of Fez (FST Fez), Morocco
Software development, algorithms, and system architecture.

**Diplôme d'Études Universitaires Scientifiques et Techniques (DEUST)** 2022 – 2024
Faculty of Sciences and Techniques of Fez (FST Fez), Morocco
Foundations in mathematics, computer science, and problem solving.

## Professional Experience

**Software Engineering Intern** 2025 | 2 months
École Polytechnique des Génies Fez, Morocco

- Designed and developed an internal academic resource management platform using Java, Spring Boot, and React.js.
- Built RESTful backend services and integrated them with a responsive user interface for organized resource access and management.
- Practiced backend architecture, API design, and frontend–backend integration in an institutional setting.

## Selected Projects – End-to-End Systems

### AI Fraud Detection & Explainable Intelligence

Completed – Local Demonstration

**Stack:** Python, Scikit-learn, SHAP, FastAPI, SQLite, Next.js, TypeScript, Tailwind CSS

Developed an end-to-end fraud analytics demonstration using a historical anonymized credit-card dataset, designed to support interpretable and risk-prioritized transaction review.

- Combined a calibrated Logistic Regression classifier with threshold-based decisions and an independent Isolation Forest anomaly signal.
- Integrated global and local SHAP explanations, policy-based risk tiers, and alert prioritization for analyst review.
- Built FastAPI prediction and CSV analysis workflows, SQLite-backed history and investigations, and a responsive Next.js dashboard.
- Evaluated models on historical held-out data; results do not represent live production performance. The local API is not authenticated and the platform is not presented as production-deployed.

GitHub repository

### Customer Churn Prediction

Completed

**Stack:** Python, Pandas, NumPy, Scikit-learn, SHAP, FastAPI, Pydantic, Pytest, Docker

Built an end-to-end customer churn prediction system using the IBM Telco dataset of 7,043 customers.

- Performed exploratory analysis, preprocessing, feature engineering, and comparison of Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models.
- Selected a Logistic Regression pipeline with a 0.30 decision threshold to prioritize identifying customers at risk; achieved a held-out ROC-AUC of 0.8429 and detected 285 of 374 churners in the test set.
- Added model-coefficient and SHAP interpretation, validated FastAPI prediction services, and a responsive

customer-input interface.
- Validated Docker execution, 39 automated tests, and 70 functional/technical checks.

GitHub repository

### Football Intelligence & Player Recommendation System

Completed

**Stack:** Python, Pandas, NumPy, Scikit-learn, FastAPI, Pydantic, React, TypeScript, Vite, Recharts

Created a football analytics and decision-support platform for player profiling, similarity search, scouting, and recruitment shortlists.

- Prepared player and event data, constructed position-aware features, per-90 statistics, and comparative percentile profiles.
- Implemented normalized statistical similarity using cosine distance, same-position recommendations, weighted scouting criteria, replacement candidates, and hidden-gem discovery.
- Delivered typed REST services and an interactive React dashboard with player comparisons, radar visualizations, and scouting workflows.
- Performed backend, API, data-validation, and frontend checks.

GitHub repository

### University Material Resource Management System

Portfolio Project

**Stack:** Java, Spring Boot, Spring Security, Spring Data JPA, PostgreSQL, REST APIs

Designed an enterprise-oriented university hardware resource management platform covering procurement, supplier proposals, inventory, assignments, and maintenance.

- Modeled workflows involving teachers, department heads, resource managers, suppliers, and technicians, with role-based access controls.
- Supported departmental needs collection, tender management, supplier offers, resource inventory and assignment, and failure reporting.
- Structured backend functionality into controller, service, and persistence layers with secured REST endpoints.

Backend repository

### Water Potability Prediction System

Machine Learning Application

**Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Joblib

Developed a classification workflow for estimating water potability from nine physicochemical measurements.

- Compared median imputation with missing-row removal and evaluated Random Forest, KNN, and SVM models.
- Prioritized recall for the non-potable class to examine the risk of unsafe water being classified as safe; reviewed confusion matrices, F1, ROC-AUC, and cross-validation.
- Created a Streamlit interface for manual inputs, predictions, sample testing, and CSV-exportable prediction history.

## Additional Software & Data Projects

### Academic Resource Management Platform

Full-Stack Application

**Stack:** Java, Spring Boot, REST API, React.js

Created an educational platform where professors can organize and upload course materials and students can securely access and download resources. Demonstrates authentication, API integration, and React-based user interfaces.

### Employee & Salary Management System

Enterprise Web Application

**Stack:** Spring Boot, Thymeleaf, JDBC, MySQL

Built a three-tier MVC application supporting employee, supervisor, and salary management, with CRUD workflows and role-based authentication for director, supervisor, and employee roles.

### Advanced Ticket Management System

In Development

**Stack:** Java, Spring Boot, Spring Data JPA, PostgreSQL, REST API

Developing a ticket management backend with ticket lifecycle states (OPEN, IN_PROGRESS, RESOLVED, CLOSED), Controller–Service–Repository separation, DTO/Mapper patterns, persistence, and Jakarta Validation.
GitHub repository

### NoSQL & Machine Learning Football Prediction System

Data Engineering / ML

**Stack:** Python, Redis, Machine Learning, REST API, React.js

Integrated Redis-backed data handling, model training, prediction endpoints, and a React frontend in a football-oriented machine learning application. GitHub repository

### Personal Portfolio Website

Published Web Portfolio

**Stack:** Next.js, React, TypeScript, Tailwind CSS, Framer Motion, EmailJS

Designed a responsive portfolio with animated navigation and landing experience, project pages, skills, a professional timeline, certification gallery, and contact integration. Organized project data into reusable components and centralized records. Live website

GitHub repository

### Indonesia Tourism Website

Frontend Project

**Stack:** HTML, CSS, JavaScript, PHP

Built a responsive tourism presentation website with destination highlights, cultural content, image gallery, contact section, and interactive frontend elements. GitHub repository

## Technical Skills

**Backend & Software Engineering:** Java, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, JDBC, Maven, REST API design, layered architectures, authentication and role-based access.
**Frontend:** React, Next.js, TypeScript, JavaScript, HTML, CSS, Tailwind CSS, Bootstrap.
**Data Science & Machine Learning:** Python, Pandas, NumPy, SQL, Scikit-learn, Matplotlib, Seaborn, Jupyter, EDA, data preprocessing, feature engineering, classification, regression, clustering, model evaluation, cross-validation, threshold tuning, SHAP.
**ML Applications & APIs:** FastAPI, Pydantic, Streamlit, model serialization, prediction APIs, Pytest.
**Databases & Tools:** PostgreSQL, MySQL, Redis, SQLite, Git, GitHub, Docker, Postman.
**Additional topics listed in portfolio:** Deep Learning, TensorFlow/Keras, reinforcement learning, Spark/PySpark, MLflow, DVC, orchestration, microservices, and Spring Cloud (familiarity level to be confirmed; not presented as production experience).

## Languages & Interests

**Languages:** Arabic (native), French (good), English (basic).
**Interests:** Sports, music.

## Certifications

- **Python for Data Science, AI & Development** – IBM / Coursera (July 2026)
- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI / Coursera (July 2026)
- **Introduction to Software Engineering** – IBM / Coursera (July 2026)
- **CS250: Python for Data Scientists** – Saylor Academy
- **The Data Science Profession** – University of London / Coursera (July 2026)
- **Spring – Ecosystem and Core** – LearnQuest / Coursera (July 2026)
- **Introduction to Machine Learning** – Duke University / Coursera (July 2026)
- **Java Explorer** – Oracle (LinkedIn credential post)
- **Python for Machine Learning** – Simplilearn
- **Data Visualization** – Kaggle